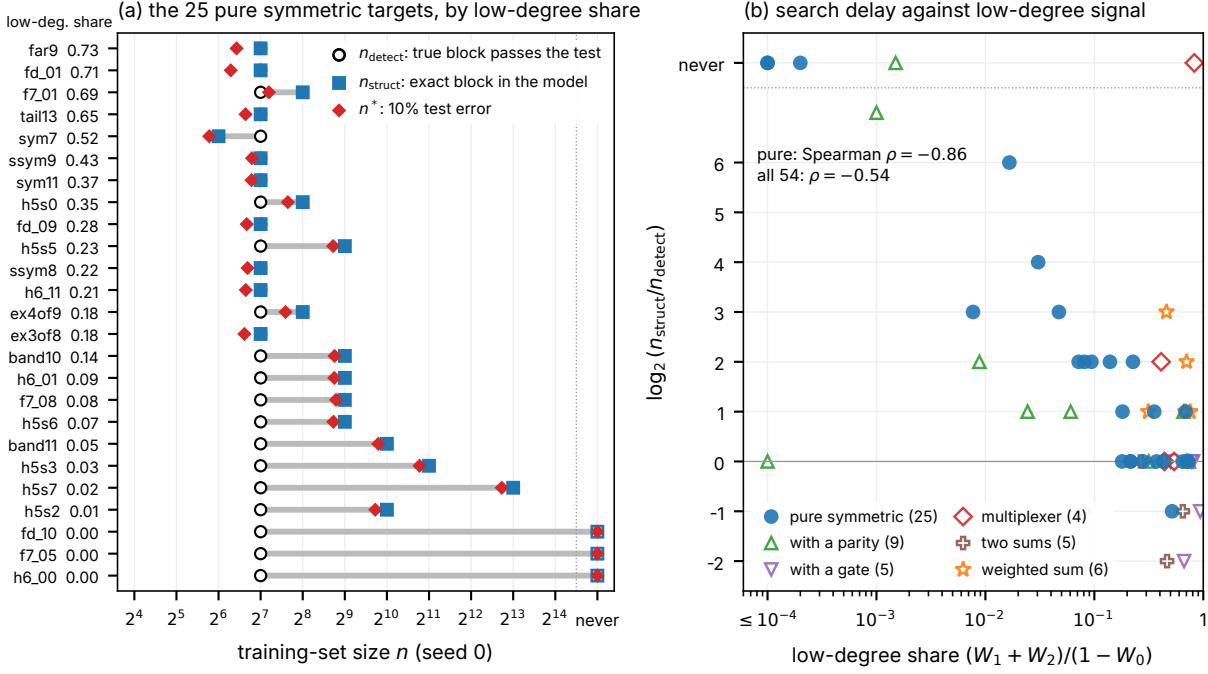
(a) the 25 pure symmetric targets, by low-degree share
low-deg. share
far9 0.73
fd_01 0.71
f7_01 0.69
tail13 0.65
sym7 0.52
ssym9 0.43
sym11 0.37
h5s0 0.35
fd_09 0.28
h5s5 0.23
ssym8 0.22
h6_11 0.21
ex4of9 0.18
ex3of8 0.18
band10 0.14
h6_01 0.09
f7_08 0.08
h5s6 0.07
band11 0.05
h5s3 0.03
h5s7 0.02
h5s2 0.01
fd_10 0.00
f7_05 0.00
h6_00 0.00
n_{detect}: true block passes the test
n_{struct}: exact block in the model
n^*: 10% test error
2^4 2^5 2^6 2^7 2^8 2^9 2^{10} 2^{11} 2^{12} 2^{13} 2^{14} never
training-set size n (seed 0)
(b) search delay against low-degree signal
never
6
5
4
3
2
1
0
-1
-2
$\log_2(n_{struct}/n_{detect})$
pure: Spearman $\rho = -0.86$
all 54: $\rho = -0.54$
pure symmetric (25)
with a parity (9)
with a gate (5)
multiplexer (4)
two sums (5)
weighted sum (6)
$\leq 10^{-4}$ 10^{-3} 10^{-2} 10^{-1} 1
low-degree share $(W_1 + W_2)/(1 - W_0)$